# Artificial Intelligence
# Advanced Topics in AI & ML
## Introduction. Course logistics and syllabus

Aleksandr Petiushko

ML Research



# Content

1. Introduction

# Content

1. Introduction
2. Course logistics and syllabus

# Intro

## About the lecturer[1]

- Aleksandr Petiushko, PhD in theoretical CS (2016)
- Lecturer in Lomonosov MSU / MIPT for Machine Learning, Computer Vision, Deep Learning Theory, Python for an ML Researcher since 2019
- Former Huawei Chief Scientist (Scientific Expert), AIRI Director of Key Research Programs (Leading Scientific Researcher), Nuro Head of ML Research, Gatik Head of AI Research, Logical Intelligence Head of AI



[1] Homepage: `https://petiushko.info/`

# Intro

Time to introduce yourselves: what are your hobbies, motivation in ML, etc.: please go into "**Module 1 Students Introduction**" thread

# Sofia Plagiarism Policy

- It covers parts "*sourced from AI*"
  - Please read the "**Sofia Plagiarism Policy**" thread
  - **First offense**: students need to rewrite assignment
  - **Second offense**: students fail the course
  - **Third offense**: students re to be withdrawn from their program

# Note about xGPTx

- It can produce very plausible answers in 90% of cases

# Note about xGPTx

- It can produce very plausible answers in 90% of cases
- The caveats are the following:

# Note about xGPTx

- It can produce very plausible answers in 90% of cases
- The caveats are the following:
  - It can really hallucinate some things which are just untrue

# Note about xGPTx

- It can produce very plausible answers in 90% of cases
- The caveats are the following:
  - It can really hallucinate some things which are just untrue
  - It can produce very different information in comparison to the source used to ask question (e.g., book chapter)

# Course logistics

- Course grading will be done based on attendance, assignments, discussions, mini research problem, coding tasks, and the final exam.

# Course logistics

- Course grading will be done based on attendance, assignments, discussions, mini research problem, coding tasks, and the final exam.
- Contribution:
  - 58%: attendance, assignments, discussions
  - 21%: MRP
  - 21%: final exam

# Course logistics

- Course grading will be done based on attendance, assignments, discussions, mini research problem, coding tasks, and the final exam.
- Contribution:
  - 58%: attendance, assignments, discussions
  - 21%: MRP
  - 21%: final exam
- Preliminary grading scale:

| Grade | Percent accumulated |
|---|---|
| A | $\geq$ 90 % |
| B | 80-89 % |
| C | 70-79 % |

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)
- MRP to be done in form of a pdf presentation that should be shared via student's github

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)
- MRP to be done in form of a pdf presentation that should be shared via student's github
- It should contain:
  - *tl;dr*: motivation, main idea, and results briefly (1 slide)

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)
- MRP to be done in form of a pdf presentation that should be shared via student's github
- It should contain:
  - *tl;dr*: motivation, main idea, and results briefly (1 slide)
  - Literature review on the related methods

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)
- MRP to be done in form of a pdf presentation that should be shared via student's github
- It should contain:
  - *tl;dr*: motivation, main idea, and results briefly (1 slide)
  - Literature review on the related methods
  - Approach used (with all relevant links used)

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)
- MRP to be done in form of a pdf presentation that should be shared via student's github
- It should contain:
  - *tl;dr*: motivation, main idea, and results briefly (1 slide)
  - Literature review on the related methods
  - Approach used (with all relevant links used)
  - Link to the code used to achieve the goal

# Note about optional MRP

- MRP = **M**ini **R**esearch **P**roblem
- Topics: all the directions learned through the lectures (but not limited to - just contact your instructor in advance)
- MRP to be done in form of a pdf presentation that should be shared via student's github
- It should contain:
  - *tl;dr*: motivation, main idea, and results briefly (1 slide)
  - Literature review on the related methods
  - Approach used (with all relevant links used)
  - Link to the code used to achieve the goal
- Presentations to be presented by the author during the last onground session
  - Duration: apprx 10 min

# Late Submission Policy

Late submission deduction percent: **15% every day**;

# Late Submission Policy

Late submission deduction percent: **15% every day**;

- It means that if you're **7 days late** you'll get **0 score** but **need to submit anyway**.

# Makeup Policy

Missing onground class:

- Student's grade is dropped by 10%;

# Makeup Policy

Missing onground class:

- Student's grade is dropped by 10%;

Missing Exam:

- A student gets 0 for the Exam;

# Makeup Policy

Missing onground class:

- Student's grade is dropped by 10%;

Missing Exam:

- A student gets 0 for the Exam;

Missing Assignment in time (1 week, please refer to the *Late Submission Policy*):

- No makeups, i.e. 0 for the Assignment;

# Makeup Policy

Missing onground class:

- Student's grade is dropped by 10%;

Missing Exam:

- A student gets 0 for the Exam;

Missing Assignment in time (1 week, please refer to the *Late Submission Policy*):

- No makeups, i.e. 0 for the Assignment;

Unless:

- A Student has a serious medical condition, and this condition is validated by a hospital or licensed California physician (in English)
- The Student contacts in time our chair (Samane Abdi) and Professor and describes the situation and provides all the needed proofs

# Github

- Course page: `https://github.com/fatheral/ai-intro-course`
- The professor's lectures are uploaded there

# Takeaway notes

1. Please go through all the materials of **Module 0**

# Thank you!